DeadlineDesk — UML Class Diagram (Week 7 implementation)

Classes and functions as they exist in code/core. Models extend django.db.models.Model (User extends AbstractUser); views, services and decorators are module-level functions, shown as «module» classifiers.

core.models (Django ORM, stored in SQLite)

«Django» AbstractUser
username, password, email,
first_name, last_name, is_active, …

Company
+name: str «unique»
+description: str [blank]
+created_at: datetime
+__str__(): str

User
+role: Role = STUDENT
+roll_number: str [blank]
+__str__(): str

Assignment
+title: str
+description: str
+due_at: datetime
+late_policy: LatePolicy
+grace_minutes: int = 0
+penalty_percent: int (0–100) = 0
+created_at: datetime
+__str__(): str

PlacementRound
+title: str
+opens_at: datetime
+closes_at: datetime
+status: Status = DRAFT
+created_at: datetime
+effective_status: Status «property»
+effective_status_label: str «property»
+__str__(): str

Submission
+file: FileField (media/submissions/)
+note: str [blank]
+submitted_at: datetime = now
+late_status: LateStatus
+penalty_percent: int = 0
+similarity_score: Decimal(5,2) = 0
+status: Status = SUBMITTED
«unique» (assignment, student)
+__str__(): str

ReminderLog
+scheduled_for: datetime
+status: Status = SCHEDULED
+sent_at: datetime [0..1]
+created_at: datetime

AuditLog
+action: str
+entity_type: str
+entity_id: int
+detail: dict (JSON)
+created_at: datetime
+action_label: str «property»

Grade
+score: Decimal ≥ 0
+max_score: Decimal ≥ 1 = 100
+feedback: str [blank]
+graded_at: datetime
+__str__(): str

ChecklistItem
+title: str
+required: bool = True
+position: int = 0
+__str__(): str

ChecklistCompletion
+completed_at: datetime
«unique» (item, student)

«enumeration» User.Role
STUDENT
TA
PLACEMENT_ADMIN

«enumeration» PlacementRound.Status
DRAFT
PUBLISHED
OPEN
CLOSED
ARCHIVED

«enumeration» ReminderLog.Status
SCHEDULED
SENT

«enumeration» Assignment.LatePolicy
REJECT
PENALTY
GRACE

«enumeration» Submission.Status
SUBMITTED
UNDER_REVIEW
GRADED

«enumeration» Submission.LateStatus
ON_TIME
GRACE
PENALTY

Application layer (code/core)

«module» core.views
+home(request)
+dashboard(request) [login]
+placement_list(request) [login]
+company_create(request) [PLACEMENT_ADMIN]
+round_create(request) [PLACEMENT_ADMIN]
+round_detail(request, pk) [login]
+checklist_create(request, pk) [PLACEMENT_ADMIN]
+round_publish(request, pk) [PLACEMENT_ADMIN, POST]
+checklist_toggle(request, pk) [STUDENT, POST]
+assignment_list(request) [login]
+assignment_create(request) [TA]
+assignment_detail(request, pk) [login]
+assignment_submit(request, pk) [STUDENT]
+submission_detail(request, pk) [login]
+submission_download(request, pk) [login]
+submission_review(request, pk) [TA, POST]
+submission_grade(request, pk) [TA]
-_audit(actor, action, instance, detail)

«module» core.decorators
+role_required(*roles)
login_required, then 403
if user.role is not allowed

«module» core.forms
CompanyForm
PlacementRoundForm +clean()
ChecklistItemForm
AssignmentForm +clean()
SubmissionForm +clean_file()
GradeForm +clean()
DateTimeLocalInput
Each Meta.model is the
model of the same name.

«call»

«module» core.services
+evaluate_lateness(*, due_at,
submitted_at, policy, grace_minutes=0,
penalty_percent=0): LateDecision
+similarity_stub(uploaded_file,
comparison_submissions): Decimal
-_tokens(text): set[str]

«create»

«dataclass, frozen» LateDecision
+accepted: bool
+status: str | None
+penalty_percent: int
+reason: str

«management command» seed_demo.Command
extends BaseCommand
+handle(): demo users, round,
checklist, reminder, assignment

Solid arrow: association from the class holding the foreign key (role = field name, with multiplicities). Hollow triangle: inheritance. Dashed arrow: dependency; unlabelled ones are «use» (views use the decorators, forms and models; services read Assignment and Submission). [login] = @login_required; [ROLE] = @role_required(ROLE). Not drawn: core.admin registrations and Django's built-in LoginView/LogoutView.